

Lambda-RAG — AI That Stands Up in Court

A document review platform that uses AI to *write* the rules — then enforces them with pure code. Same document + same rules = same verdict. Every time. Forever.



Reproducible

Byte-identical results across runs — 35 regression tests prove it on every code change



Auditable

Every verdict cites the exact rule, the exact policy clause, and the exact paragraph in the document



Defensible

No AI in the decision loop — verdicts survive regulatory challenge, subpoena, and tribunal review

Where AI Helps — and Where It Steps Aside

The LLM is a brilliant author. It's a terrible judge. We use it for one, not the other.

AUTHORING — ONCE, OFFLINE, AI-ASSISTED



Feed it your policy or regulation



AI extracts candidate rules (temp=0, structured)



Human expert reviews & approves each rule



Rules are signed, fingerprinted, version-locked



Signed Ruleset — the ONLY thing that crosses →

⚡ RUNTIME — EVERY REVIEW, DETERMINISTIC, NO AI



Document arrives (contract, permit, design...)



Parsed & projected into structured categories



Each rule's selector finds matching sections



Lambda expression evaluates: Pass / Fail / Gap



Verdict report + redlined Word doc emitted

How Projection Rules Work — A Real Example

Plain-language policy becomes a testable, repeatable rule applied over any document paragraph.

① THE POLICY SAYS

"Undisputed invoices shall be payable within **thirty (30) days** of receipt of a valid invoice."

— Treasury Board Directive on Payments

② AI DRAFTS → HUMAN APPROVES → RULE

NATURAL LANGUAGE DESCRIPTION:

"Payment terms must be net 30 days or fewer from receipt of an undisputed invoice."

SELECTOR (WHICH PARAGRAPHS TO CHECK):

```
category == "payment_terms"
```

LAMBDA EXPRESSION (THE TEST):

```
text.Contains("30 days")  
|| text.Contains("net 30")
```

```
|| text.Contains("15 days")
```

③ CONTRACT PARAGRAPH UNDER REVIEW

"5.2 Payment. Customer shall remit payment within **net 30** calendar days following receipt of Vendor's undisputed invoice."

✓ PASS

④ A DIFFERENT CONTRACT...

"5.2 Payment. All amounts shall be due **net 60 days** from date of invoice."

✗ FAIL — Remediation provided

→ "Replace with: Customer shall pay all undisputed invoices within thirty (30) days."

Why This Beats "Just Ask the AI"

For anything where the answer has to survive a lawyer, auditor, or regulator.

What Matters	LLM Alone	Lambda-RAG
Run it twice — same answer?	✗ Different each time	✓ Byte-identical, proven by tests
"Show me why."	✗ Cites can hallucinate	✓ Rule ID + policy clause + exact paragraph
Reproduce in 18 months?	✗ Model versions change	✓ Fingerprinted ruleset, any time, forever
Reorder paragraphs — verdict change?	✗ LLMs are order-sensitive	✓ Each rule evaluates independently
"The document is silent on this."	✗ Invents plausible answer	✓ Returns GAP — silence ≠ compliance
Speed of authoring?	✓ Fast for one-off	✓ AI writes rules once, applies thousands of times

Key Concepts — The Novel Pattern

How prose becomes enforceable code.

Projection Rule

A machine-testable rule derived from a regulation or policy. Contains a *selector* (what to look at), a *lambda* (what to check), and a *citation* (where this requirement comes from). Authored with AI help, enforced without it.

Lambda Expression

A small, deterministic test — like a formula in a spreadsheet — that returns true or false. "Does this paragraph mention 30 days?" is a lambda. It runs the same way, with the same result, every time.

Topic Map Projection

The step that turns unstructured paragraphs into categorized buckets (e.g., "this paragraph is about payment terms"). Uses a fixed dictionary — not AI — so the same document always maps the same way.

Signed Ruleset

The locked, fingerprinted bundle of all rules for a domain. Once signed, it cannot drift. If the fingerprint matches, the verdict is reproducible — guaranteed by the system, provable by the tests.

Gap Verdict

When a rule applies but the document says nothing about it. Other systems either guess or silently pass. Lambda-RAG flags it explicitly — because silence is not the same as compliance.

Golden-Master Tests

35 automated proofs that run on every change, across 5 industries. They guarantee that the engine hasn't drifted — the same inputs still produce the same verdicts, byte for byte.

The bet: AI is the best authoring tool we've ever had. It's the worst possible judge for anything that has to be defended.

Lambda-RAG — Open Source · .NET 9 · 5 Industry Verticals · 35 Golden-Master Proofs
github.com/MTCMarkFranco/lambda-rag